

Figure 1: Histograms of forgetting events on (from left to right) *MNIST*, *permutedMNIST* and *CIFAR-10*. Insets show the zoomed-in y-axis.

This network achieves a competitive 3.99% test error. For full experimentation details, see the Supplementary.

4 CHARACTERIZING EXAMPLE FORGETTING

Number of forgetting events We estimate the number of forgetting events of all the training examples for the three different datasets (*MNIST*, *permutedMNIST* and *CIFAR-10*) across 5 random seeds. The histograms of forgetting events computed from one seed are shown in Figure 1. There are 55,012, 45,181 and 15,628 unforgettable examples common across 5 seeds, they represent respectively 91.7%, 75.3%, and 31.3% of the corresponding training sets. Note that datasets with less complexity and diversity of examples, such as *MNIST*, seem to contain significantly more unforgettable examples. *permutedMNIST* exhibits a complexity balanced between *MNIST* (easiest) and *CIFAR-10* (hardest). This finding seems to suggest a correlation between forgetting statistics and the intrinsic dimension of the learning problem, as recently formalized by Li et al. (2018).

Stability across seeds To test the stability of our metric with respect to the variance generated by stochastic gradient descent, we compute the number of forgetting events per example for 10 different random seeds and measure their correlation. From one seed to another, the average Pearson correlation is 89.2%. When randomly splitting the 10 different seeds into two sets of 5, the cumulated number of forgetting events within those two sets shows a high correlation of 97.6%. We also ran the original experiment on 100 seeds to devise 95% confidence bounds on the average (over 5 seeds) number of forgetting events per example (see Appendix 13). The confidence interval of the least forgotten examples is tight, confirming that examples with a small number of forgetting events can be ranked confidently.

Forgetting by chance In order to quantify the possibility of forgetting occurring by chance, we additionally analyze the distribution of forgetting events obtained under the regime of random update steps instead of the true SGD steps. In order to maintain the statistics of the random updates similar to those encountered during SGD, random updates are obtained by shuffling the gradients produced by standard SGD on a main network (more details are provided in Appendix 12). We report the histogram of chance forgetting events in Supplementary Figure 13: examples are being forgotten by chance a small number of time, at most twice and most of the time less than once. The observed stability across seeds, low number of chance forgetting events and the tight confidence bounds suggest that it is unlikely for the ordering produced by the metric to be the by-product of another unrelated random cause.

Algorithm 1 Computing forgetting statistics.

```

initialize prev_acci = 0, i ∈ D
initialize forgetting T[i] = 0, i ∈ D
while not training done do
    B ∼ D # sample a minibatch
    for example i ∈ B do
        compute acci
        if prev_acci > acci then
            T[i] = T[i] + 1
        prev_acci = acci
    gradient update classifier on B
return T

```



Figure 2: Pictures of unforgettable (*Top*) and forgettable examples (*Bottom*) of every *CIFAR-10* class. Forgettable examples seem to exhibit peculiar or uncommon features. Additional examples are available in Supplemental Figure 15.

First learning events We investigate whether unforgettable and forgettable examples need to be presented different numbers of times in order to be learnt for the first time (*i.e.* for the first learning event to occur, as defined in Section 3). The distributions of the presentation numbers at which first learning events occur across all datasets can be seen in Supplemental Figure 8. We observe that, while both unforgettable and forgettable sets contain many examples that are learnt during the first 3-4 presentations, the forgettable examples contain a larger number of examples that are first learnt later in training. The Spearman rank correlation between the first learning event presentations and the number of forgetting events across all training examples is 0.56, indicating a moderate relationship.

Misclassification margin The definition of forgetting events is binary and as such fairly crude compared to more sophisticated estimators of example relevance (Zhao & Zhang, 2015; Chang et al., 2017). In order to qualify its validity, we compute the misclassification margin of forgetting events. The misclassification margin of an example is defined as the mean classification margin (defined in Section 3) over all its forgetting events, a negative quantity by definition. The Spearman rank correlation between an example’s number of forgetting events and its mean misclassification margin is -0.74 (computed over 5 seeds, see corresponding 2D-histogram in Supplemental Figure 9). These results suggest that examples which are frequently forgotten have a large misclassification margin.

Visual inspection We visualize some of the unforgettable examples in Figure 2 along with some examples that have been most forgotten in the *CIFAR-10* dataset. Unforgettable samples are easily recognizable and contain the most obvious class attributes or centered objects, *e.g.*, a plane on a clear sky. On the other hand, the most forgotten examples exhibit more ambiguous characteristics (as in the center image, a truck on a brown background) that may not align with the learning signal common to other examples from the same class.

Detection of noisy examples We further investigate the observation that the most forgettable examples seem to exhibit atypical characteristics. We would expect that if highly forgettable examples have atypical class characteristics, then noisily-labeled examples will undergo more forgetting events. We randomly change the labels of 20% of *CIFAR-10* and record the number of forgetting events of both the noisy and regular examples through training. The distributions of forgetting events across noisy and regular examples are shown in Figure 3. We observe that the most forgotten examples are those with noisy labels and that no noisy examples are unforgettable. We also compare the forgetting events of the noisy examples to that of the same set of examples with original labels and observe a much higher degree of forgetting in the noisy case. The results of these synthetic experiments support the hypothesis that highly forgettable examples exhibit atypical class characteristics.

4.1 CONTINUAL LEARNING SETUP

We observed that in harder tasks such as *CIFAR-10*, a significant portion of examples are forgotten at least once during learning. This leads us to believe that catastrophic forgetting may be observed, to some extent, even when considering examples coming from the same task distribution. To test this hypothesis, we perform an experiment inspired by the standard continual learning setup (McCloskey & Cohen, 1989; Kirkpatrick et al., 2017). We create two tasks by randomly sampling 10k examples

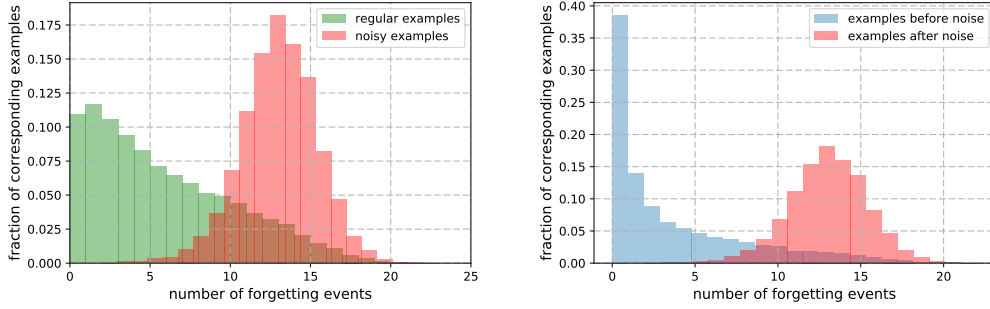


Figure 3: Distributions of forgetting events across training examples in *CIFAR-10* when 20% of labels are randomly changed. *Left*. Comparison of forgetting events between examples with noisy and original labels. The most forgotten examples are those with noisy labels. No noisy examples are unforgettable. *Right*. Comparison of forgetting events between examples with noisy labels and the same examples with original labels. Examples exhibit more forgetting when their labels are changed.

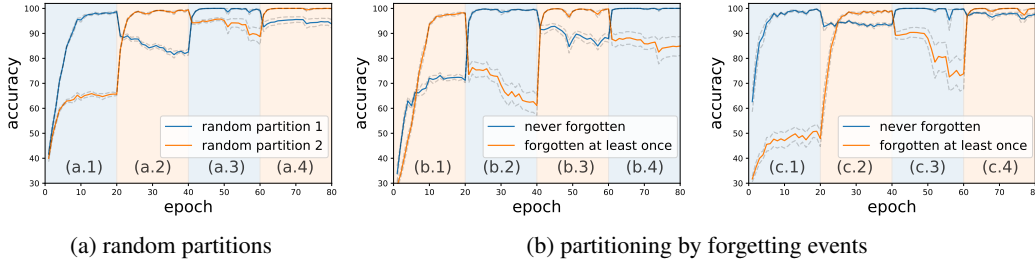


Figure 4: Synthetic continual learning setup for *CIFAR-10*. Background color in each column indicates the training partition, curves track performance on both partitions during interleaved training. Solids lines represent the average of 5 runs and dashed lines represent the standard error. The figure highlights that examples that have been forgotten at least once can “support” those that have never been forgotten, as shown in (c.2) and (b.3).

from the *CIFAR-10* training set and dividing them in two equally-sized partitions (5k examples each). We treat each partition as a separate “task” even though they should follow the same distribution. We then train a classifier for 20 epochs on each partition in an alternating fashion, while tracking performance on both partitions. The results are reported in Figure 4 (a). The background color represents which of the two partitions is currently used for training. We observe some forgetting of the second task when we only train on the first task (panel (a.2)). This is somewhat surprising as the two tasks contain examples from the same underlying distribution.

We contrast the results from training on random partitions of examples with ones obtained by partitioning the examples based on forgetting statistics (Figure 4 (b)). That is, we first compute the forgetting events for all examples based on Algorithm 1 and we create our tasks by sampling 5k examples that have zero forgetting events (named *f0*) and 5k examples that have non-zero forgetting events (named *fN*). We observe that examples that have been forgotten at least once suffer a more drastic form of forgetting than those included in a random split (compare (a.2) with (b.2)). In panel (b.3) and (c.2) we can observe that examples from task *f0* suffer very mild forgetting when training on task *fN*. This suggests that examples that have been forgotten at least once may be able to “support” those that have never been forgotten. We observe the same pattern when we investigate the opposite alternating sequence of tasks in Figure 4 (b, right).